

1 Main components

Figure 1 shows the model, Figure 2 the models across settings, and Figure 3 the per-region comparison. Tables 1, 2, 3, 4, and 5 give the numbers.

2 Supplementary components

Figure 4 and Table 6 show the field separation, Figure 5 the static development ladder, and Table 7 the structural ablations.

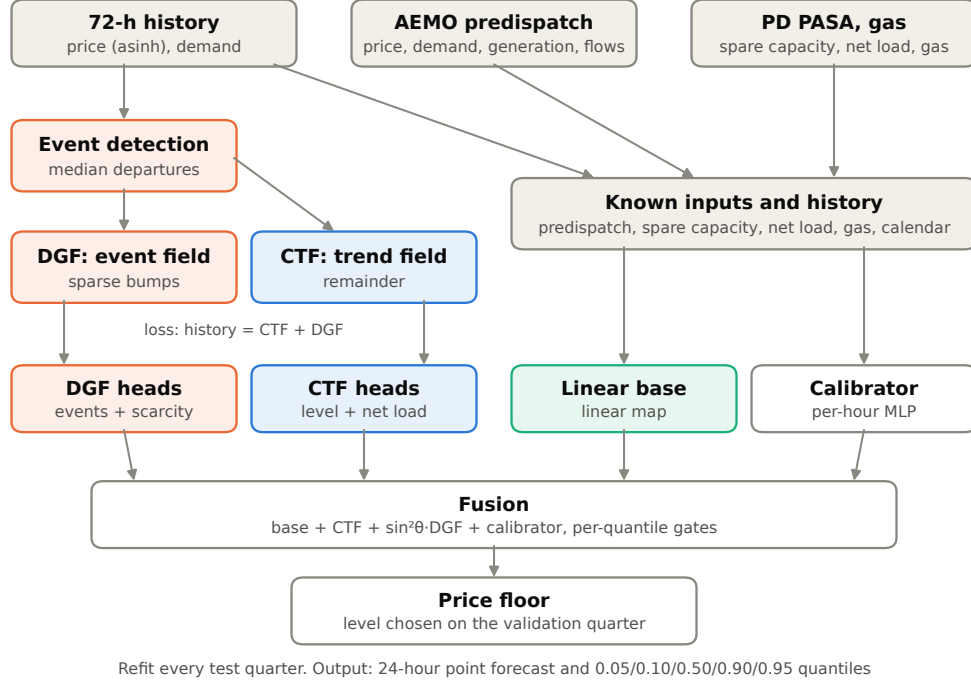
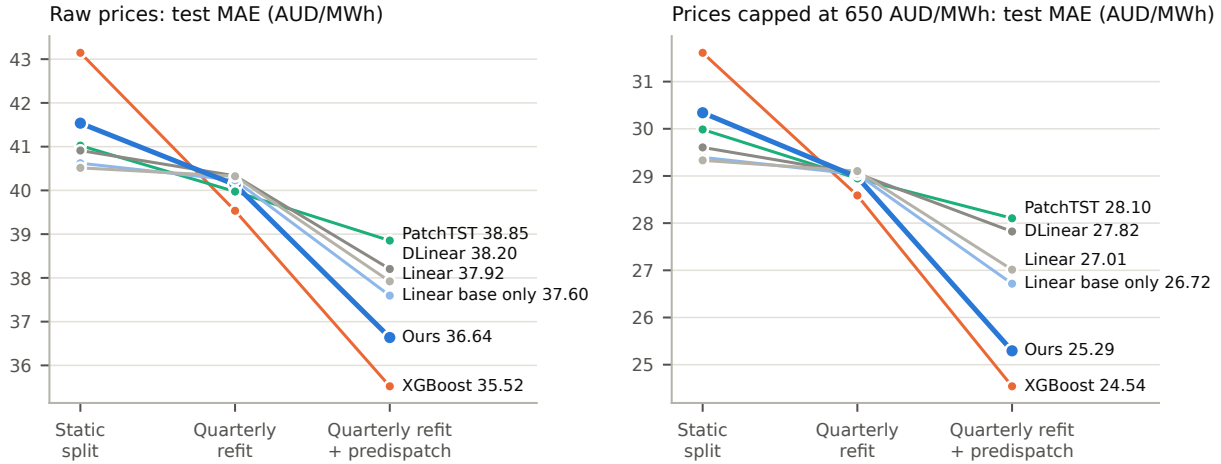


Figure 1: Dual-field forecaster with a linear base. Events detected in the 72-hour history, as the largest departures from the window median after non-maximum suppression, form the discrete event field (DGF); the continuous trend field (CTF) fits the remainder under the decomposition loss. Field-specific linear heads and a linear base on the raw history and all known inputs (AEMO predispatch, PD PASA spare capacity and net load, gas price, calendar) are fused by a trigonometric gate; a per-hour calibrator corrects the response to the known inputs, and the forecast is clipped at a price floor chosen on the validation quarter. Every model is refit before each test quarter.



Three-region means over seeds 2026-2028. Ours in the last setting includes the calibrator and the validation-chosen price floor. Axes do not start at zero.

Figure 2: Three-region mean test MAE of every model in three settings: a single fit on 2015–2021 (static split), a refit before each 2023–2024 test quarter, and a refit with AEMO predispatch inputs. On the static split the dual-field model is behind a linear model with the same inputs; with recalibration and predispatch it is ahead of every model except XGBoost. Means over seeds 2026–2028; axes do not start at zero.

Table 1: Main comparison on the 2023–2024 test set (17,521 hourly origins per region, 24-hour horizon). Every model is refit before each test quarter and uses the same inputs, including AEMO predispach and no news; cells are means over three seeds, and Mean MAE shows the seed standard deviation. Ours includes the calibrator and the price floor; Linear base only is ours without the dual fields. Window RMSE is averaged over forecast windows; 90% AIS is the average interval score of the 5–95% interval; CRPS is approximated by twice the mean pinball loss over the quantiles 0.05, 0.10, 0.50, 0.90, 0.95. Lower is better; best in bold.

Model	MAE			Mean MAE	Window RMSE	90% AIS	CRPS
	NSW	QLD	TAS				
<i>Raw prices</i>							
Ours	44.27	38.57	27.07	36.64 $\pm$ 0.14	73.69	367.8	25.07
Linear base only	44.86	39.01	28.91	37.60 $\pm$ 0.03	74.35	401.2	26.04
XGBoost	43.34	36.47	26.76	35.52 $\pm$ 0.02	73.14	347.6	23.11
PatchTST	46.68	40.03	29.84	38.85 $\pm$ 0.07	75.94	355.3	24.33
DLinear	46.09	39.54	28.98	38.20 $\pm$ 0.09	76.11	338.7	23.62
Linear	45.16	39.51	29.10	37.92 $\pm$ 0.02	74.78	339.7	23.66
<i>Capped at 650</i>							
Ours	25.64	25.89	24.35	25.29 $\pm$ 0.17	36.07	165.8	13.80
Linear base only	26.77	27.46	25.92	26.72 $\pm$ 0.00	38.08	183.8	14.96
XGBoost	25.20	24.78	23.65	24.54 $\pm$ 0.02	35.51	160.7	13.10
PatchTST	28.77	28.67	26.87	28.10 $\pm$ 0.07	40.00	188.6	15.12
DLinear	29.01	28.44	26.02	27.82 $\pm$ 0.05	41.50	185.7	14.96
Linear	27.09	27.84	26.11	27.01 $\pm$ 0.01	38.42	186.1	14.98

Table 2: Diebold–Mariano tests, ours minus comparator, in the main setting (negative favours ours). Losses are per-origin MAE averaged over seeds and per-origin CRPS of seed-averaged quantiles; Δ columns pool the three regions, and NSW, QLD, TAS give the raw-price MAE difference per region. Newey–West variance with 48 lags. * $p < 0.05$, ** $p < 0.01$.

Comparator	Raw prices					Capped at 650	
	Δ MAE	NSW	QLD	TAS	Δ CRPS	Δ MAE	Δ CRPS
Linear base only	-0.96**	-0.59	-0.44	-1.84**	-1.42**	-1.42**	-1.18**
XGBoost	+1.11**	+0.93**	+2.10**	+0.31	+1.11**	+0.75**	+0.31**
PatchTST	-2.22**	-2.41**	-1.46**	-2.77**	-0.20	-2.81**	-1.69**
DLinear	-1.57**	-1.82*	-0.97*	-1.91**	+0.58	-2.53**	-1.46**
Linear	-1.28**	-0.89*	-0.93**	-2.03**	+0.53	-1.72**	-1.49**

Table 3: When the dual fields help, and what each later component adds. Pooled Diebold–Mariano differences (negative is better); fields are compared with the same linear base in each setting, and each later row with the row it modifies. * $p < 0.05$, ** $p < 0.01$.

Change	Raw prices		Capped at 650	
	Δ MAE	Δ CRPS	Δ MAE	Δ CRPS
<i>Fields minus linear base</i>				
Static split	+0.92**	-1.90**	+0.95**	-1.88**
Quarterly refit	-0.10	-1.28**	-0.06	-1.06**
Quarterly refit + predispach	-0.34*	-1.13**	-0.70**	-0.94**
<i>Added to fields + predispach</i>				
Hinge features on predispach price	-0.25**	-0.95*	-0.39**	-0.26**
Calibrator	-0.31*	-0.26*	-0.44**	-0.22**
Price floor (after calibrator)	-0.31**	-0.03**	-0.29**	-0.01
Interconnector inputs (after floor)	+0.04	+0.04	+0.19**	-0.04

Table 4: Three-region mean test MAE across settings (static split, quarterly refit, quarterly refit with predispatch), means over three seeds. Ours in the last setting includes the calibrator and the price floor.

Model	Raw prices			Capped at 650		
	Static	Refit	Refit + PD	Static	Refit	Refit + PD
Ours	41.54	40.13	36.64	30.34	28.98	25.29
Linear base only	40.62	40.23	37.60	29.39	29.04	26.72
XGBoost	43.15	39.53	35.52	31.61	28.59	24.54
PatchTST	41.02	39.97	38.85	29.99	28.95	28.10
DLinear	40.91	40.33	38.20	29.61	29.05	27.82
Linear	40.52	40.32	37.92	29.33	29.10	27.01

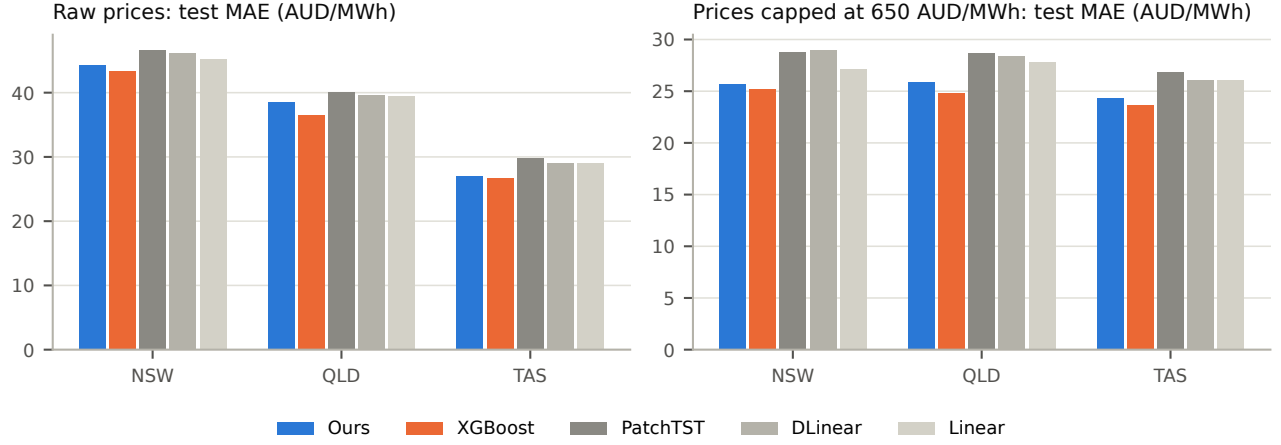


Figure 3: Test MAE per region in the main setting (quarterly refit with predispatch inputs) on raw prices (left) and prices capped at 650 AUD/MWh (right). All models share the splits, inputs, refit schedule, and scoring; bars are means over three seeds.

Table 5: Probabilistic scores in the price tails, main setting, mean of the three regions, quantiles averaged over seeds: CRPS and the 0.05-quantile pinball loss on hours with negative prices, CRPS and the 0.95-quantile pinball loss on spike hours (> 300 AUD/MWh), and the coverage of the 90% interval. Lower is better; best in bold. $^{\dagger}/^{\ddagger}$: ours is significantly better/worse than that model (pooled Diebold–Mariano test on per-origin event-hour losses, $p < 0.05$).

Model	Negative-price hours		Spike hours (> 300)		
	CRPS	$q_{0.05}$ pinball	CRPS	$q_{0.95}$ pinball	90% coverage
<i>Raw prices</i>					
Ours	14.52	4.74	800.3	675.3	91.5%
Linear base only	16.21 †	5.82 †	882.3 †	777.2 †	93.1%
XGBoost	15.96 †	6.13 †	790.5	640.3	87.7%
PatchTST	17.99 †	7.51 †	745.2 ‡	600.3 ‡	90.0%
DLinear	17.25 †	7.33 ‡	740.3‡	591.7‡	89.1%
Linear	17.25 †	7.38 ‡	741.8 ‡	593.8 ‡	89.0%
<i>Capped at 650</i>					
Ours	14.28	4.83	121.1	77.2	90.5%
Linear base only	16.23 †	6.07 †	146.7 †	87.5 †	92.4%
XGBoost	15.47 †	6.13 †	138.3 †	91.0 †	86.2%
PatchTST	17.90 †	7.42 †	127.8	77.0	90.0%
DLinear	17.19 †	7.32 ‡	141.7 †	85.5	89.0%
Linear	17.31 †	7.44 ‡	141.9 †	85.5	88.7%

Table 6: Share of past spike-hour prices (> 300 AUD/MWh) carried by the event field on the test split, transplanted Gabor DGF against the detected-event DGF.

Event field	NSW	QLD	TAS
Transplanted (Gabor)	-0.03	-0.00	-0.04
Detected events (ours)	0.33	0.33	0.28

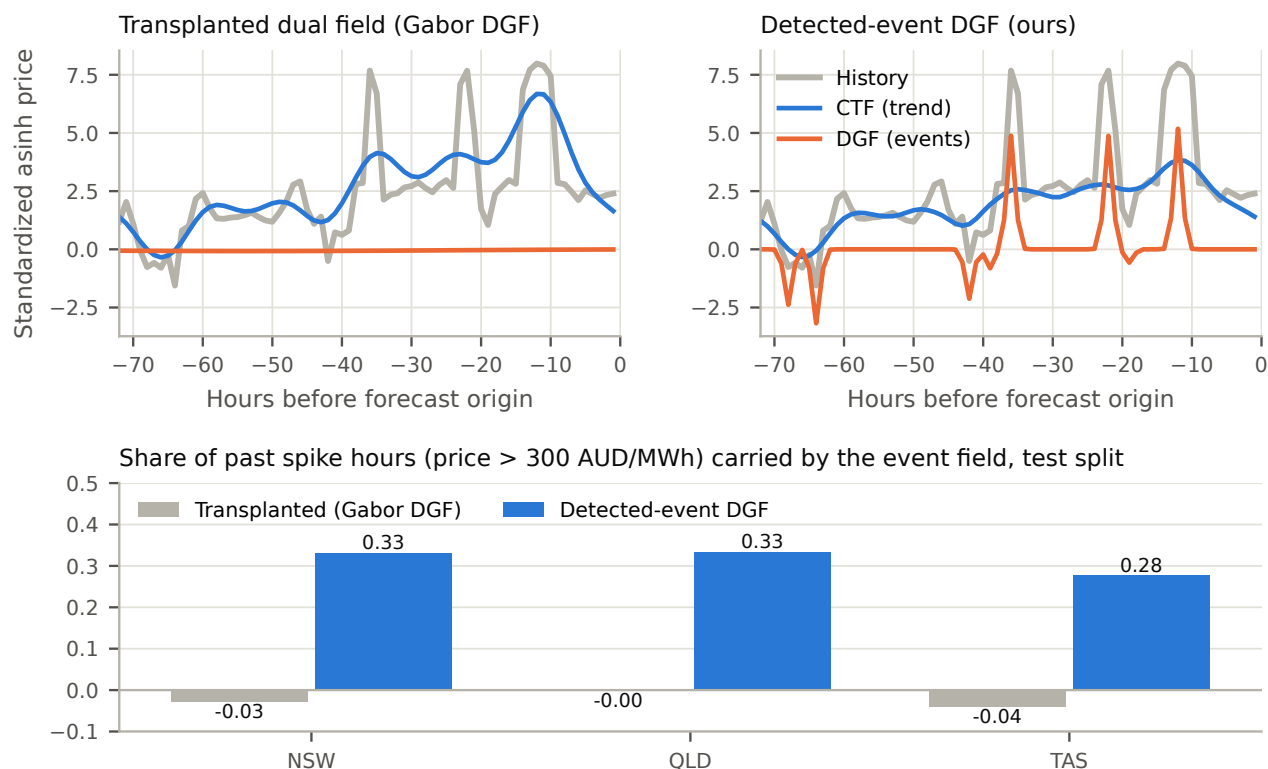


Figure 4: Field separation. *Top*: one NSW test window in standardized asinh price. Transplanted from reconstruction, the Gabor DGF stays flat and the CTF absorbs the spikes (left); with detected events the spikes and troughs move into the DGF and the CTF keeps the level (right). *Bottom*: share of past spike-hour prices (> 300 AUD/MWh) carried by the event field on the test split: -0.04 to 0.00 for the transplanted model, 0.28 to 0.33 for ours.

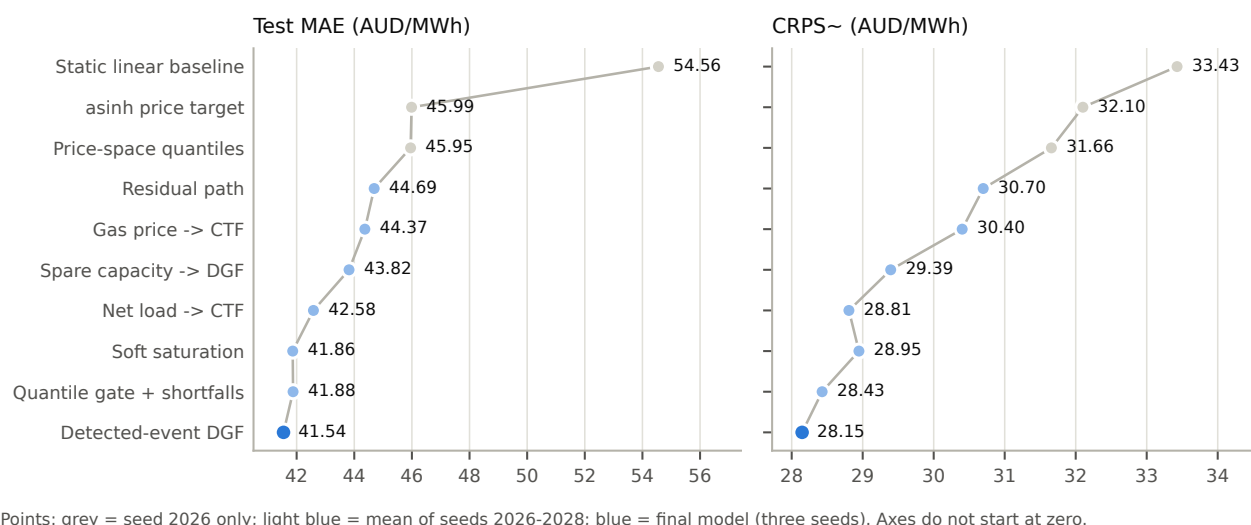


Figure 5: Development ladder on the static split: three-region mean test MAE and CRPS on raw prices. Each step changes one component of the step above. Grey points are seed 2026 only; blue points are means over seeds 2026-2028. Axes do not start at zero.

Table 7: Structural ablations of the detected-event DGF on the static split, raw prices, three-region means over seeds 2026–2028 (MAE $\pm$ seed standard deviation). The last column gives the paired MAE difference to the detected-event DGF for each seed; positive means the variant is worse.

Variant	MAE	Window RMSE	CRPS	Δ MAE vs detected (seeds)
<i>Structure</i>				
Transplanted dual field (Gabor DGF)	41.88 $\pm$ 0.22	79.21	28.43	+0.38 / +0.03 / +0.58
No decomposition (heads read raw history)	41.90 $\pm$ 0.11	79.31	28.40	+0.30 / +0.25 / +0.51
Adaptive attention atoms + echo	41.84 $\pm$ 0.05	79.01	28.09	+0.30 / +0.28 / +0.30
Detected events + daily echo	42.21 $\pm$ 0.11	79.93	28.22	+0.79 / +0.47 / +0.73
Detected-event DGF	41.54 $\pm$ 0.06	78.80	28.15	–
<i>Detector settings (8 events, 3-hour separation, learned threshold, median)</i>				
4 events	41.65 $\pm$ 0.03	78.85	28.25	+0.13 / +0.01 / +0.18
16 events	41.50 $\pm$ 0.08	78.76	28.15	-0.08 / -0.03 / -0.03
No suppression (1-hour separation)	41.85 $\pm$ 0.03	79.00	28.05	+0.36 / +0.21 / +0.35
6-hour separation	41.51 $\pm$ 0.05	78.77	28.28	-0.05 / -0.11 / +0.05
No threshold	41.63 $\pm$ 0.08	78.98	27.97	+0.21 / -0.03 / +0.07
Mean baseline	41.64 $\pm$ 0.03	78.91	28.02	+0.09 / +0.05 / +0.15